2025 WASSCE MATHEMATICS 2 SOLUTION

QUESTION	DETAILS	MARKS
1. (a)	$t_1 = \frac{150}{60} = 2.5hrs$	M1 for either time
		A1 for 2.5
	$t_2 = \frac{150}{90} = \frac{5}{3}hrs$	A1 for $\frac{5}{3}$ (accept 1.67)
	$t_{total} = 2.5 + \frac{5}{3} = \frac{25}{6}hrs$	B1 for $\frac{25}{6}$ or its equivalent
	Average speed $= 300 \div \frac{25}{6}$	M1 for finding average speed
	$= 72km/hr$	A1 for $72km/hr$
(b)	New time for 1st half $= \frac{150}{50} = 3hrs$	B1 for 3hrs
	Change in time = 3 – 2.5	
	= 0.5hrs	
	Overall time will increase by 0.5hrs or 30mins.	B1 overall time will increase by 0.5hrs or 30mins (-1 ou/wu)
		[8 Marks]
2.	Let x be amount invested at 10%	
	$1.4x$ = amount invested at 11%	B1 for $1.4x$
	Interest at 10% $= x \times \frac{10}{100} \times \frac{7}{2}$	M1 for either of the interest
	$= 0.35x$	A1 for $0.35x$
	Interest at 11% $= (1.4x) \times \frac{11}{100} \times \frac{7}{2}$	
	$= 0.539x$	A1 for $0.539x$
	$0.35x + 0.539x = 22{,}580.60$	M1 for equating interest
	$0.889x = 22{,}580.60$	M1 for solving
	$x = 25{,}400.00$	A1 for 25,400.00
	Amount invested at 11% $= 1.4 \times 25{,}400$	
	= ₦35,560.00	A1 for ₦35,560.00 (-1 ou/wu)
		[8 Marks]

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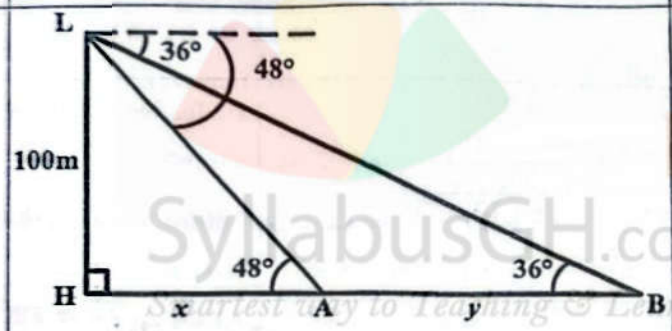
QUESTION	DETAILS	MARKS
3.	$V = \frac{1}{2} \times \frac{4}{3} \times \frac{22}{7} \times 2^3$	M1 for finding volume
	$= 16.76m^3$	A1 for $16.76m^3$
	$= 16.76 \times 1000$	M1 for 16.76×1000
	$= 16{,}760$ litres	A1 for 16,760 litres
	Water required $= 50 \times 500$	M1 for 50×500
	$= 25{,}000$ litres	A1 for 25,000 litres
	$16{,}760$ litres $< 25{,}000$ litres	B1 for the mathematical statement
	Full tank of water is not sufficient to irrigate the farm	B1 for the conclusion
		(-1 ou/wu)
		[8 Marks]
4. (a)		B2 $\left(-\frac{1}{2}ee\right)$ (36°, 48°, 100m and 90° at their respective positions)
(b)	Let $\lvert HA \rvert = x$ and $\lvert AB \rvert = y$	
	From ΔAHL	
	$\tan 48 = \frac{100}{x}$	M1 for tan ratio
	$x = \frac{100}{1.1106} = 90.0414$	A1 for 90.0414 (accept 2 d. p.)
	From ΔBHL	
	$\tan 36 = \frac{100}{x+y}$	
	$x + y = \frac{100}{0.7265} = 137.6462$	A1 for 137.6462 (accept 2 d. p.)
	$y = 137.6462 - 90.0414$	M1 for subtracting
	$= 47.6048$	A1 for 47.6048 (accept 2 d. p.)
	$= 48m$	A1 for 48m
		(-1 ou/wu) [8 Marks]

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QUESTION	DETAILS	MARKS
5. (a)	We are to find either r and s or x and y or u and v or a and b	
	$\frac{2a-4}{2} = 1$	M1 for equating either component
	$2a - 4 = 2$	M1 for solving
	$2a = 6$	
	$a = 3$	A1 for $a = 3$
	$\frac{3+b}{2} = 2a - 1$	
	$3 + b = 4a - 2$	
	$b = 4(3) - 5$	M1 for solving
	$b = 7$	A1 for $b = 7$
(b)	Either $\|YZ\|$ or $\|PH\|$ or $\|CD\|$ or $\|JK\|$	
	$Y\ (6,3),\quad Z\ (-4,7)$	B1 for both coordinates /vector correct
	$\|YZ\| = \sqrt{(6 - -4)^2 + (3-7)^2}$	M1 for use of Pythagoras
	$= 2\sqrt{29}$ units	
	$= 10.77$ units	A1 for $2\sqrt{29}$ or 10.77 units
		[8 Marks]

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QUESTION	DETAILS	MARKS
6. (a)	Total ratio = 12 + 15 + 5 = 32	
	Initial labour cost $= \frac{12}{32} \times 64{,}000$	
	$= 24{,}000.00$	B1 for 24,000.00
	Initial material cost $= \frac{15}{32} \times 64{,}000$	
	$= 30{,}000.00$	B1 for 30,000.00
	Initial contractors charges $= \frac{5}{32} \times 64{,}000$	
	$= 10{,}000.00$	B1 for 10,000.00
	The increase can be p or s or x or r	
	New labour cost $= \left(\frac{100+x}{100}\right) \times 24{,}000$	B1 for new cost of labour
	New material cost $= \left(\frac{100+2x}{100}\right) \times 30{,}000$	B1 for new cost of material
	$\left(\frac{100+x}{100}\right) \times 24{,}000 = \frac{2}{3} \times \left(\frac{100+2x}{100}\right) \times 30{,}000$	M1 for equating new cost
	$240(100+x) = 200(100+2x)$	
	$24{,}000 + 240x = 20000 + 400x$	
	$160x = 4000$	M1 for solving
	$x = 25$	A1 for 25
(b)	New labour cost $= \frac{125}{100} \times 24{,}000$	
	$= 30{,}000$	B1 for 30,000.00
	New material cost $= \frac{150}{100} \times 30{,}000$	
	$= 45{,}000$	B1 for 45,000.00
	New building cost $= 30{,}000 + 45{,}000 + 10{,}000$	M1 for adding at least 2 correct
	$= \$85{,}000.00$	A1 for $85,000.00
		(-1 ou/wu)
		[12 Marks]

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QUESTION	DETAILS	MARKS
7. (a)	The unknowns can be r or b or x or q	
	$\frac{1}{2}(3q+3)\times 7 = 98$	**M1 for finding area**
	$7(3q+3) = 196$	
	$3q+3 = 28$	**M1 for solving**
	$q = \frac{25}{3}$ or 8.3333	**A1 for $\frac{25}{3}$ or 8.3333 (accept 2 d. p)** 8.3
(b)	Area of pattern: ΔOTL or ΔPNO or ΔTRS or ΔMUV	
	Area of pattern $= \frac{1}{2}\times\frac{25}{3}\times 7$	**M1 for finding area**
	$= 29.1667$	**A1 for 29.1667 or its equivalent**
	$= 29.2m^2$	**A1 for $29.2m^2$**
(c)	We can have $\lvert RS\rvert$ or $\lvert ON\rvert$ or $\lvert TL\rvert$ or $\lvert UV\rvert$	
	$\lvert US\rvert = 3\times\frac{25}{3}+3 = 28$	**B1 for either 28 or 19.667 or its equivalent**
	$\lvert MS\rvert = 2\times\frac{25}{3}+3 = 19.667$	
	$\lvert UV\rvert = \sqrt{7^2+28^2}$	**M1 for use of Pythagoras**
	$\lvert UV\rvert = \sqrt{833}$	
	$\lvert UV\rvert = 28.8617$ or $7\sqrt{17}$	**A1 for 28.8617 or $7\sqrt{17}$ (accept 2 d. p)**
	$\lvert MV\rvert = \sqrt{7^2+19.667^2}$	
	$= \sqrt{435.7909}$	
	$= 20.8756$	**A1 for 20.8756 (accept 2 d. p)**
	Perimeter $= 8.3333 + 28.8617 + 20.8756$	**M1 for adding**
	$= 58.0706$m	
	$= 58.1$m	**A1 for 58.1m**
		(-1 ou/wu)
		[12 Marks]

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QUESTION	DETAILS	MARKS
8. (a) (i)	Let x = planned days to plough	
	Days used to plough = $x + 2$	B1 for $x + 2$
	$120x = 85(x + 2) + 40$	B1 for $120x$, B1 for adding 40 M1 for equating
	$120x = 85x + 170 + 40$ $35x = 210$	M1 for solving
	$x = 6$	A1 for 6
(ii)	Total area of farmland = 120×6	M1 for the product
	= 720acres	A1 for 720acres
(b)	$\theta = 6.8 - 5 = 1.8$	B1 for 1.8
	Distance between them = $\frac{1.8}{360} \times 2 \times \frac{22}{7} \times 6400$	M1 for finding distance
	= 201.1	A1 for 201.1
	= $201km$	A1 for 201km (-1 ou/wu)
		[12 Marks]

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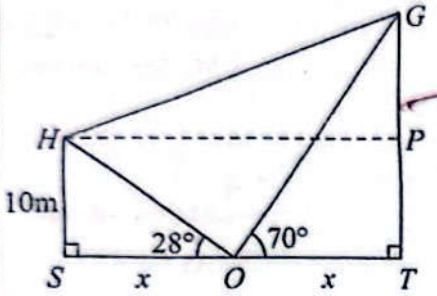
QUESTION	DETAILS	MARKS
9. (a)	Height of the pillar can be **SH** or **CR** or **XP** or **GT**. Height of the mast can be **TG** or **DP** or **YE** or **HS** with their centre as **O**.  	**B3** $\left(-\frac{1}{2}ee\right)$ **(10m, 28°, 70°, relative positions of S, H, O, G and T)**
(b) (i)	Let $\lvert OS\rvert = \lvert OT\rvert = x$ From ΔOSH $\tan 28 = \frac{10}{\lvert OS\rvert}$ $\lvert OS\rvert = \frac{10}{0.5317} = 18.8076$ From ΔOTG $\tan 70 = \frac{\lvert TG\rvert}{18.8076}$ $\lvert TG\rvert = 2.7475 \times 18.8076$ $= 51.6739$ Height of mast = 52m	 **M1 for tan ratio** **A1 for 18.8076 (accept 2 d. p.)** **M1 for tan ratio** **A1 for 51.6739 (accept 2 d. p.)** **A1 for 52m**
(ii)	From ΔHPG $\lvert PG\rvert = 51.6739 - 10$ $= 41.6739m$ $\lvert HG\rvert = \sqrt{(2 \times 18.8076)^2 + (41.6739)^2}$ $= \sqrt{1414.9033 + 1736.7139}$ $= \sqrt{3151.6172}$ $= 56.1393$ $= 56m$	 **B1 for 41.6739m (accept 2 d. p.)** **M1 for use of Pythagoras** **M1 for simplifying** **A1 for 56m** **(-1 ou/wu)** **[12 Marks]**

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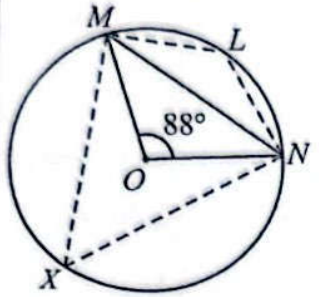
QUESTION	DETAILS	MARKS
10. (a) (i)	The points on the circle can be M, N and L or A, T and M or A, B and C or Q, R and S each with centre O.  MN OM ON 88°	B2 $\left(-\frac{1}{2}ee\right)$ (O, M, N, L and 88°)
(ii)	$\angle MXN = \frac{1}{2} \times 88$	M1 for $\frac{1}{2} \times 88$
	$= 44°$	A1 for 44°
	$\angle MLN = 180 - 44$	M1 for finding $\angle MLN$
	$= 136°$	A1 for 136°
(b) (i)	The second mason $= 1.5 \times 400$	M1 for the product
	$= 600$ blocks	A1 for 600 blocks
(ii)	First mason's rate $= \frac{400}{8}$	
	$= 50$ blocks/hr	B1 for 50 blocks/hr
	Second mason's rate $= \frac{600}{8}$	
	$= 75$ blocks/hr	B1 for 75 blocks/hr
	Total rate $= 50 + 75$	
	$= 125$	B1 for 125
	Time $= \frac{2000}{125} = 16$ hours	B1 for 16 hours
	Hence 16 hours would be required	(-1 ou/wu)
		[12 Marks

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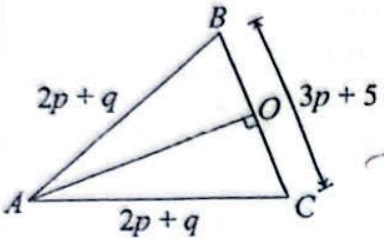
QUESTION	DETAILS	MARKS
11. (a)	The vertices of the triangle can be **ABC** with variables as p and q or **TUV** with variables as i and j or **SDR** with variables as a and b or **KLM** with variables as u and v. 	B1 $\left(-\frac{1}{2}ee\right)$ (at least 2 sides labelled)
(b) (i)	$(2p+q)+(2p+q)+(3p+5)=7q+3$	M1 for equating parameters
	$7p+2q+5=7q+3$	
	$7p+2=5q$	M1 for simplifying (all correct)
	$\frac{p}{q}=\frac{2}{3}$	
	$p=\frac{2}{3}q$	B1 for $p=\frac{2}{3}q$ or its equivalent
	$5q=7\left(\frac{2}{3}q\right)+2$	M1 for the equation
	$15q=14q+6$	
	$15q-14q=6$	M1 for solving
	$q=6$	A1 for 6
	$p=4$	A1 for 4
(ii)	$h=\sqrt{14^2-8.5^2}$	M1 for the use of Pythagoras
	$=11.1243$	A1 for 11.1243 (accept 2 d. p.)
	$\text{Area}=\frac{1}{2}\times 17\times 11.1243$	M1 for finding area
	$=94.5566$	
	$=94.6\text{km}^2$	A1 for 94.6km^2 (-1 ou/wu)
		[12 Marks]

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QUESTION	DETAILS	MARKS
12. (a)		

Class interval	Tally	f	Upper class boundary	Cumulative frequency
1 – 10	|||	3	10.5	3
11 – 20	卌	5	20.5	8
21 – 30	卌 ||	7	30.5	15
31 – 40	卌 |	6	40.5	21
41 – 50	卌 卌	10	50.5	31
51 – 60	卌 ||	7	60.5	38
61 – 70	卌	5	70.5	43
71 – 80	||||	4	80.5	47
81 – 90	||	2	90.5	49
91 – 100	|	1	100.5	50

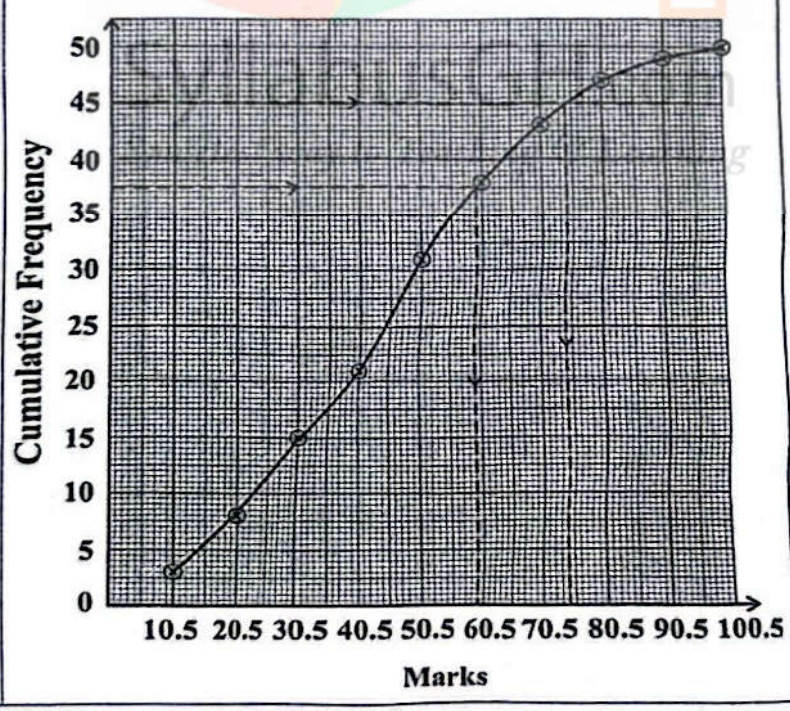
M1 for any of the two frequencies correct

A2$\left(-\frac{1}{2}ee\right)$ for all correct

M1 for first two cumulative frequencies correct

A1 for all correct

(b)



B1 for the axes (correct upper class boundary marks, labelling and CF)

B2 for CF $\left(-\frac{1}{2}ee\right)$

QUESTION	DETAILS	MARKS
(c)	Position of upper quartile $= \frac{3}{4} \times 50$ $= 37.5$ Upper quartile $= 59.5 \pm 0.5$	**M1 evidence from the graph** **A1 for 59.5 ± 0.5**
(d)	$\frac{10}{100} \times 50 = 5$ pupils passed $= 50 - 5 = 45$ Pass mark $= 75.5 \pm 0.5$	**M1 evidence from the graph** **A1 for 75.5 ± 0.5** **[12 Marks]**

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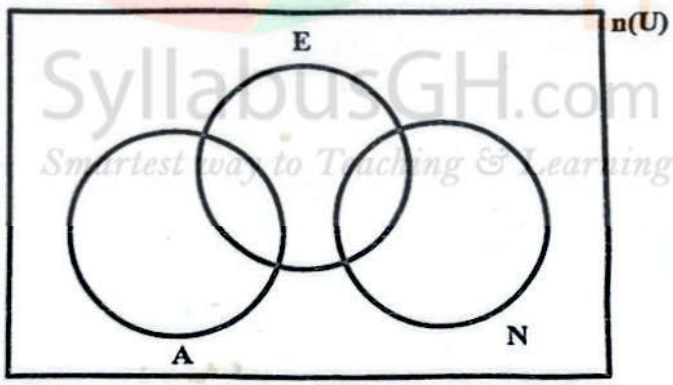
QUESTION	DETAILS	MARKS
13. (a) (i)	The required angles can be $\angle EOF$ or $\angle POQ$ or $\angle AOB$	
	$\angle GJF = 110°$	B1 for 110°
	$\angle EGF = 180 - 30 - 110$	M1 for finding $\angle EGF$
	$= 40°$	A1 for 40°
	$\angle EOF = 2 \times 40$	
	$= 80°$	B1 for 80°
(ii)	The required angles can be $\angle JEH$ or $\angle UEH$ or $\angle FAD$ or $\angle TPS$	
	$\angle EHF = \angle EGF = 40°$	B1 for $\angle EHF$
	$\angle JEH = 180 - 110 - 40$	M1 for finding $\angle JEH$
	$= 30°$	A1 for 30°
(b) (i)		B3 (-1 ee) (correct position of E, A, N and the intersections)
(ii) (α)	Not valid	B1 for Not Valid
(β)	Not valid	B1 for Not Valid
		[12 Marks]

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